

Activity 1

For the provided data about Galveston Texas in terms of the buildings polygons shapefiles and the 50yr WSE for hurricane storm surge. Calculate the flood depth at each building (Use either NOAA LiDAR imagery or USGS to download the DEM). Provide the new shapefile with extracted flood depth at each building.

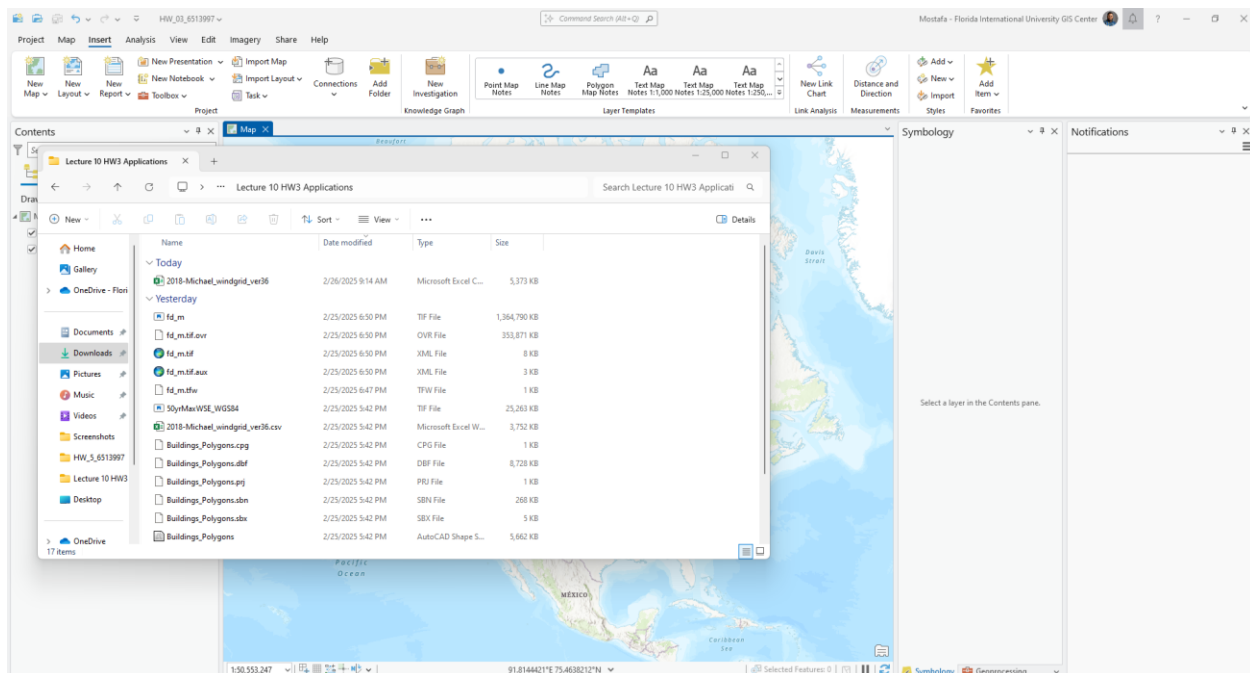
Activity 2

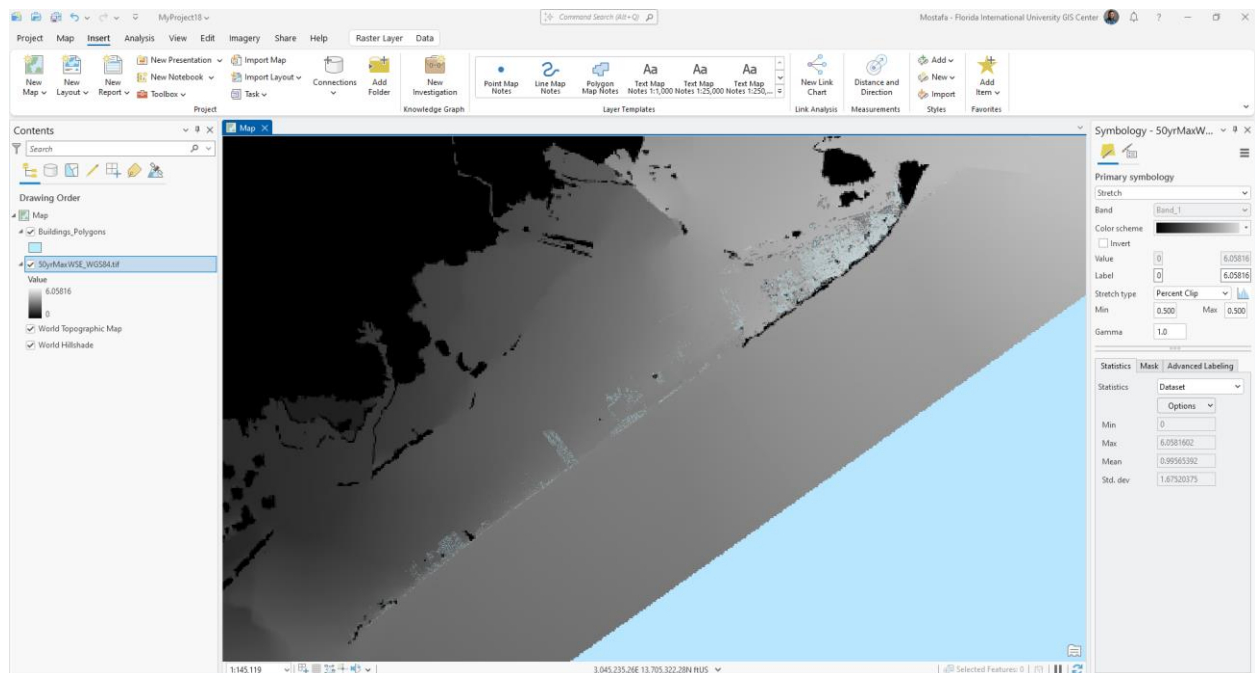
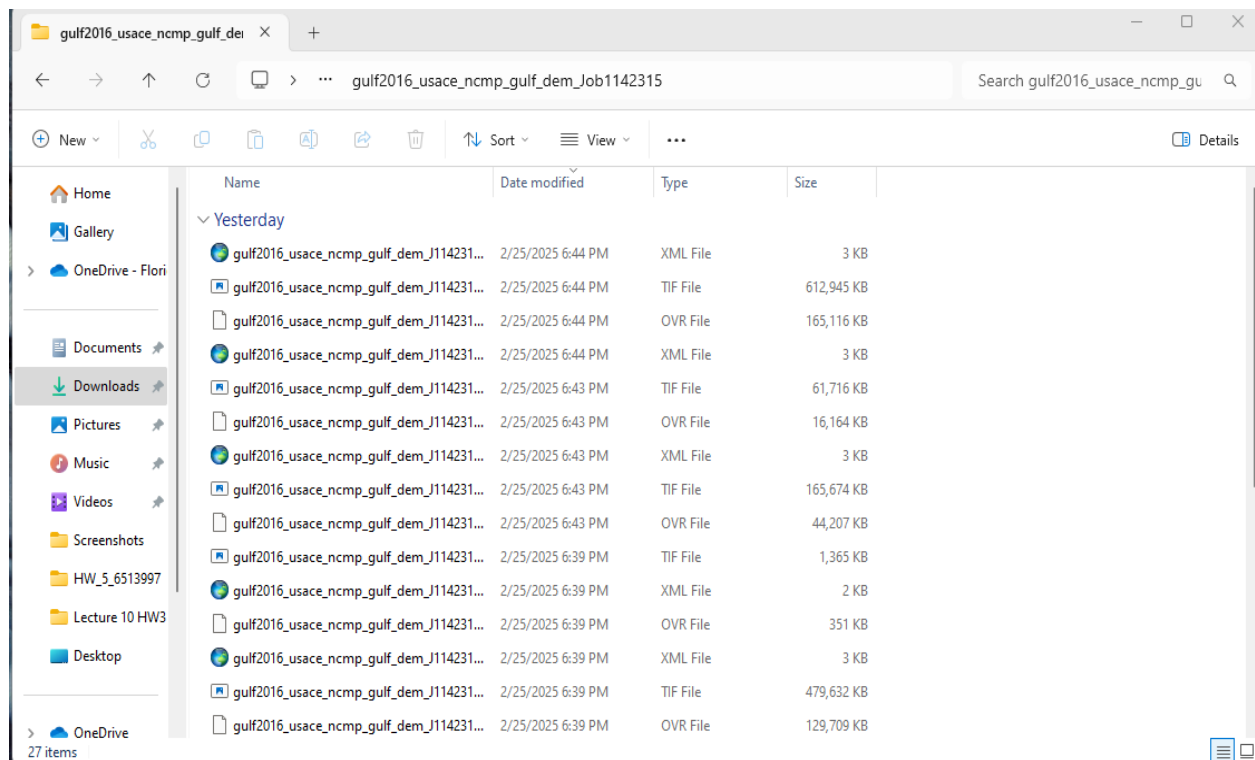
For the provided excel file for the wind grid for Hurricane Michael. Convert this grid to a raster map. Provide the raster map of the 3-sec gust wind speed.

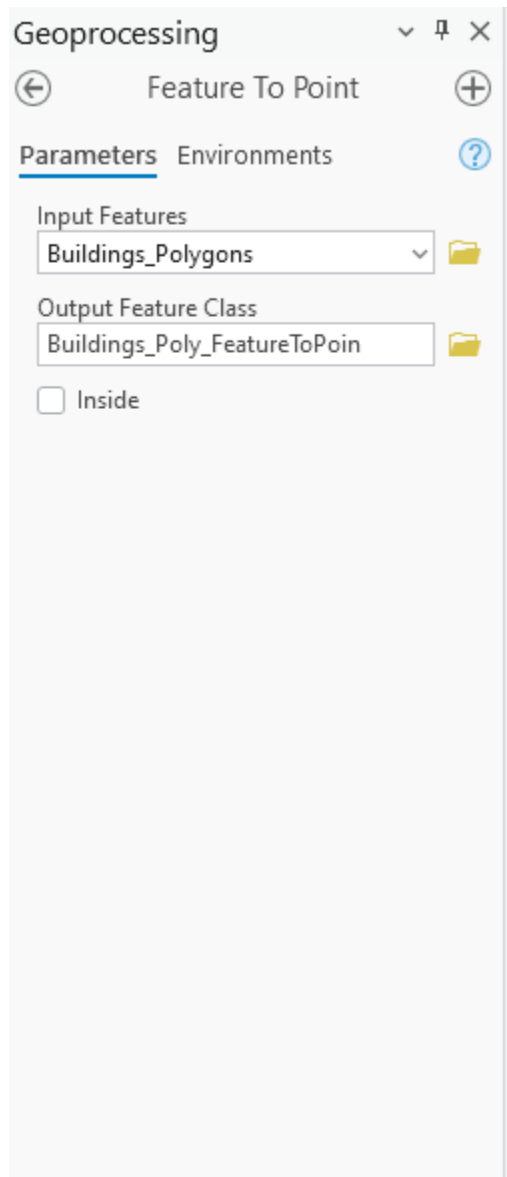
You can access the data for this assignment from here:

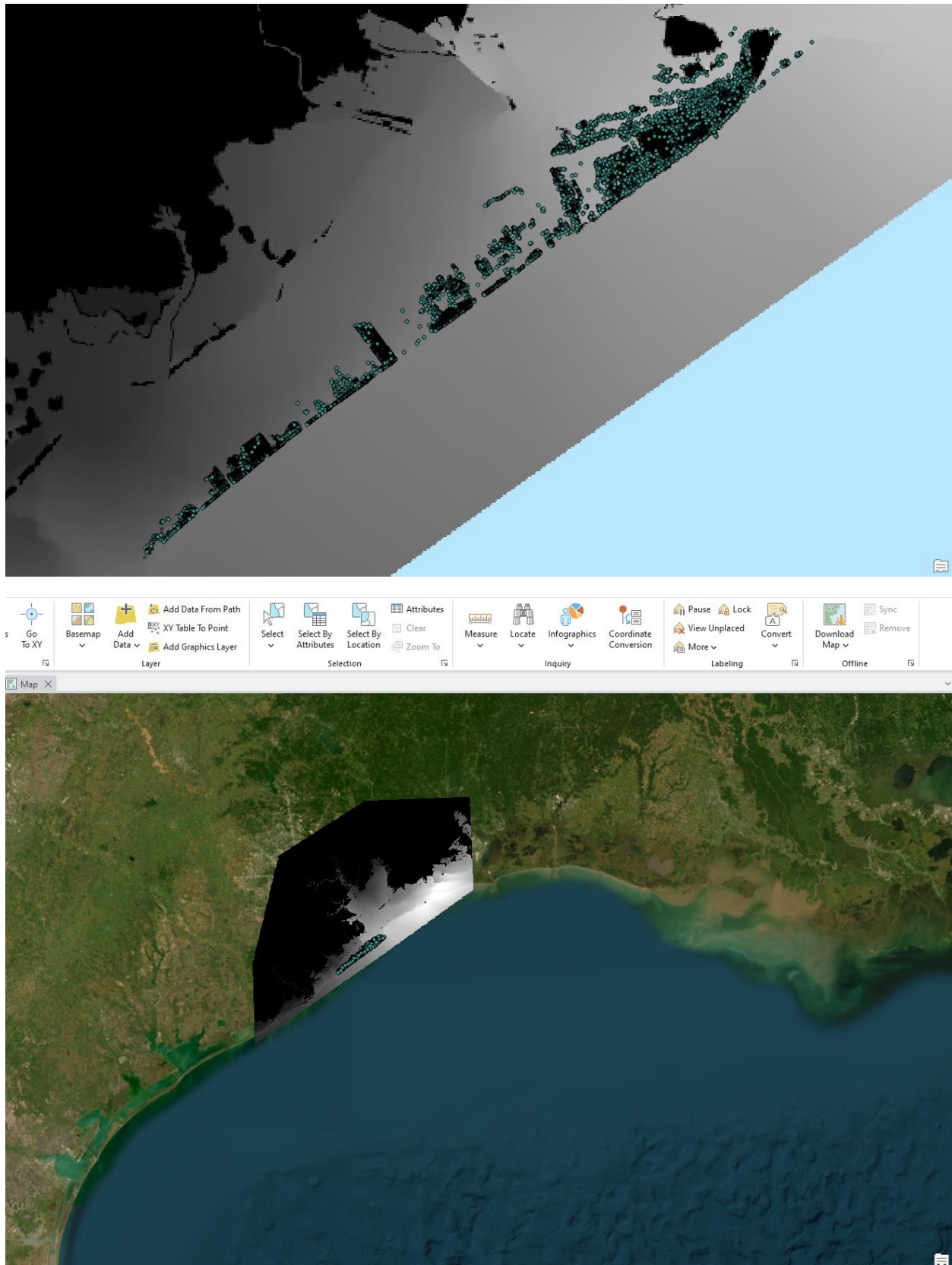
https://drive.google.com/drive/folders/1JGGGq6GVggiiS5Spex_eij5RTIJWiGmC?usp=sharing

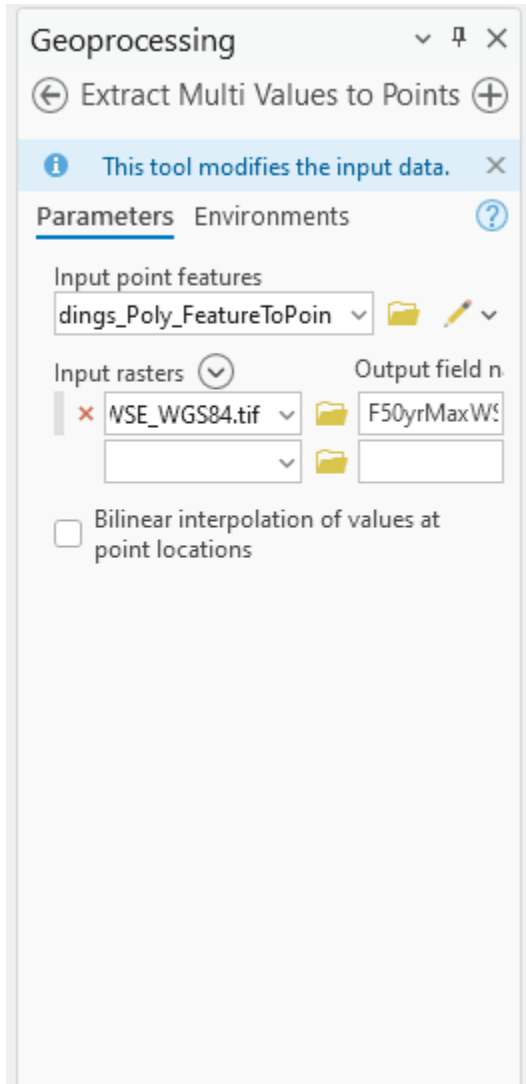
Activity 1:



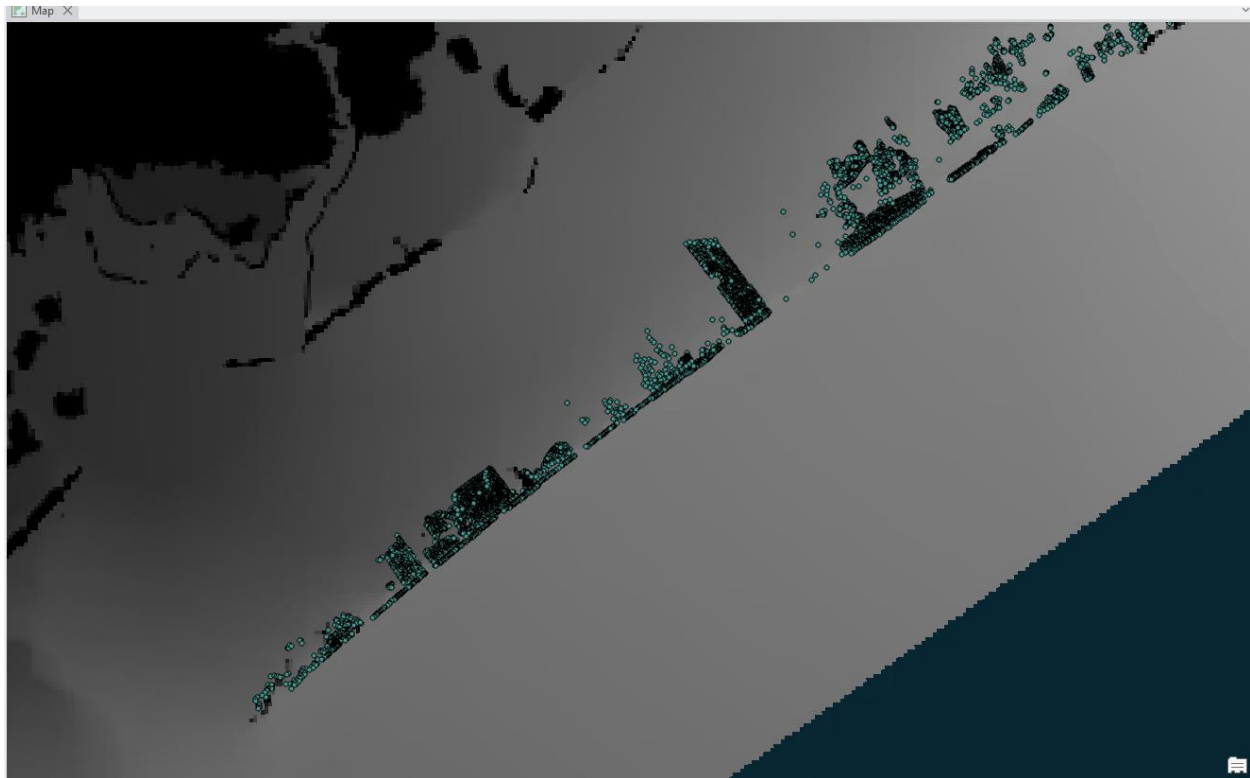




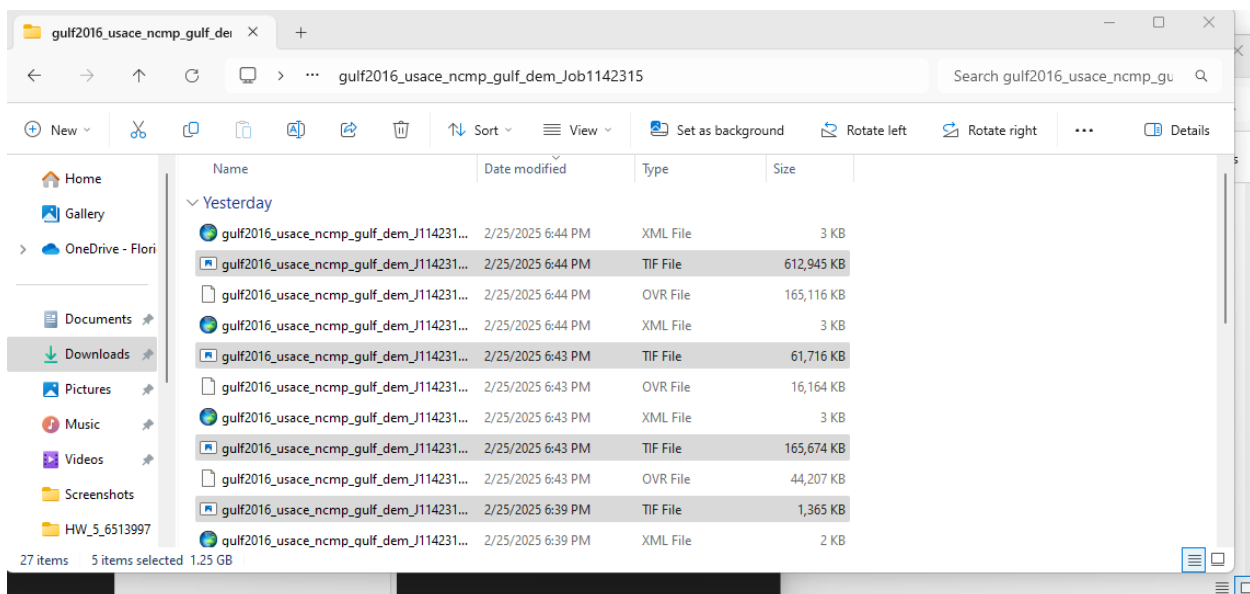




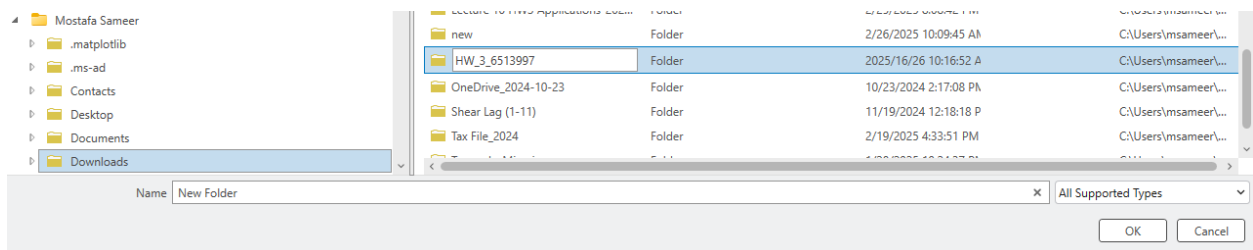
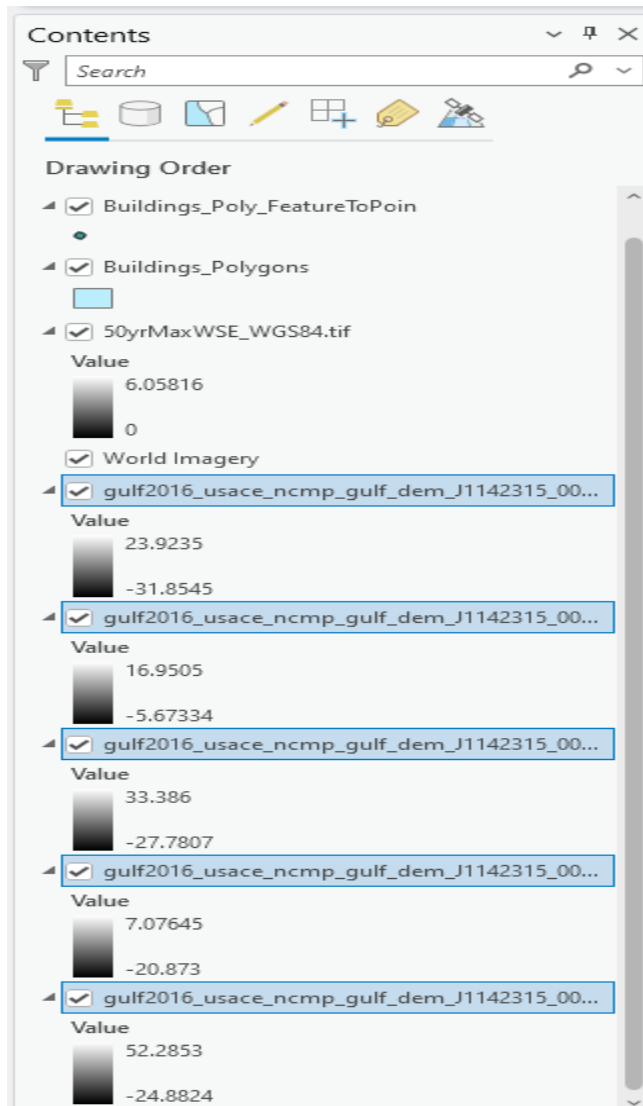
Mostafa Sameer
6513997



Saving to a new folder



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Geoprocessing

←

Mosaic To New Raster

+

Parameters

Environments

?

Input Rasters

ilf_dem_J1142315_000_000.tif

ilf_dem_J1142315_000_001.tif

ilf_dem_J1142315_001_000.tif

ilf_dem_J1142315_002_001.tif

x

ilf_dem_J1142315_001_001.tif

Output Location

HW_3_6513997

! Raster Dataset Name with Extension

Spatial Reference for Raster

Pixel Type

32 bit float

Cellsize

* Number of Bands

Mosaic Operator

Last

Mosaic Colormap Mode

First

Run

8 | Page

Geoprocessing

←

Mosaic To New Raster

+

Parameters

Environments

?

Input Rasters

ilf_dem_J1142315_000_000.tif

ilf_dem_J1142315_000_001.tif

ilf_dem_J1142315_001_000.tif

ilf_dem_J1142315_002_001.tif

×

ilf_dem_J1142315_001_001.tif

Output Location

HW_3_6513997

Raster Dataset Name with Extension

DEM.tif

Spatial Reference for Raster

NAD_1983_2011_StatePlane_Texas

Pixel Type

32 bit float

Cellsize

Number of Bands

1

Mosaic Operator

Last

Mosaic Colormap Mode

First

Run

Mosaic To New Raster

Saving raster, hit Esc to stop

9%

View Details

Open History

Suggestion

Geoprocessing

Extract Multi Values to Points

This tool modifies the input data.

Parameters

Environments

Input point features

dings_Poly_FeatureToPoin

Input rasters

Output field n

15_000_000.tif

gulf2016_usi

☐ Bilinear interpolation of values at point locations

Run

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1:1,351,528 2,225,562,49E 13,820,308.98N RUS Selected Features: 0

Buildings_Poly_FeatureToPoin *Fields: Buildings_P...y_FeatureToPoin

Current Layer Buildings_Poly_FeatureToPoin

Visible	Read Only	Field Name	Alias	Data Type	Allow NULL	Highlight	Number Format	Domain	Default	Length
☒	☒	OBJECTID_1	OBJECTID_1	Object ID	☐	☐	Numeric			
☒	☐	Shape	Shape	Geometry	☒	☐				
☒	☐	OBJECTID	OBJECTID	Long	☒	☐	Numeric			
☒	☐	Parcel_ID	Parcel_ID	Text	☒	☐				254
☒	☐	Shape_Leng	Shape_Leng	Double	☒	☐	Numeric			
☒	☐	ORIG_FID	ORIG_FID	Long	☒	☐	Numeric			
☒	☐	F50yrMaxWSE_WGS84	F50yrMaxWSE_WGS84	Double	☒	☐	Numeric			
☒	☐	gulf2016_usace_ncmp_gulf_dem_J1142315_000_000	gulf2016_usace_ncmp_gulf_dem_J1142315_000_000	Float	☒	☐	Numeric			
☒	☐	FD_ft	FD_m	Double	☒	☐	Numeric			
☒	☐	Ele_m		Double	☒	☐	Numeric			

Calculate Field

This tool modifies the Input Table

Input Table: Buildings_Poly_FeatureToPoin

Field Name (Existing or New): FD_m

Expression Type: Python

Expression

Fields: OBJECTID, Parcel_ID, Shape_Leng, ORIG_FID, F50yrMaxWSE_WGS84, gulf2016_usace_ncmp_gulf_, FD_m, Ele_m

Helpers: .as_integer_ratio(), .capitalize(), .center(), .conjugate(), .count(), .decode(), .denominator()

Insert Values: *

FD_ft =

```
!F50yrMaxWSE_WGS84! - !  
gulf2016_usace_ncmp_gulf_dem_J1142315_000_000!
```

Code Block

Enable Undo ☐

Apply OK

Calculate Field ? x

This tool modifies the Input Table x

Input Table
Buildings_Poly_FeatureToPoin

Field Name (Existing or New)
Ele_m

Expression Type
Python

Expression

Fields

- Parcel_ID
- Shape_Leng
- ORIG_FID
- F50yrMaxWSE_WGS84
- gulf2016_usace_ncmp_gulf_
- FD_m
- Ele_m

Helpers

- .as_integer_ratio()
- .capitalize()
- .center()
- .conjugate()
- .count()
- .decode()
- .denominator()

Insert Values

Ele_m =

!FD_ft!*0.3048

Code Block

Enable Undo ☐ Apply OK

1:1,351,528 1,621,363.92E 13,843,773.01N ftUS Selected Features: 0

Buildings_Poly_FeatureToPoin

Field: Add Calculate Selection: Select By Attributes Zoom To Switch Clear Delete Copy

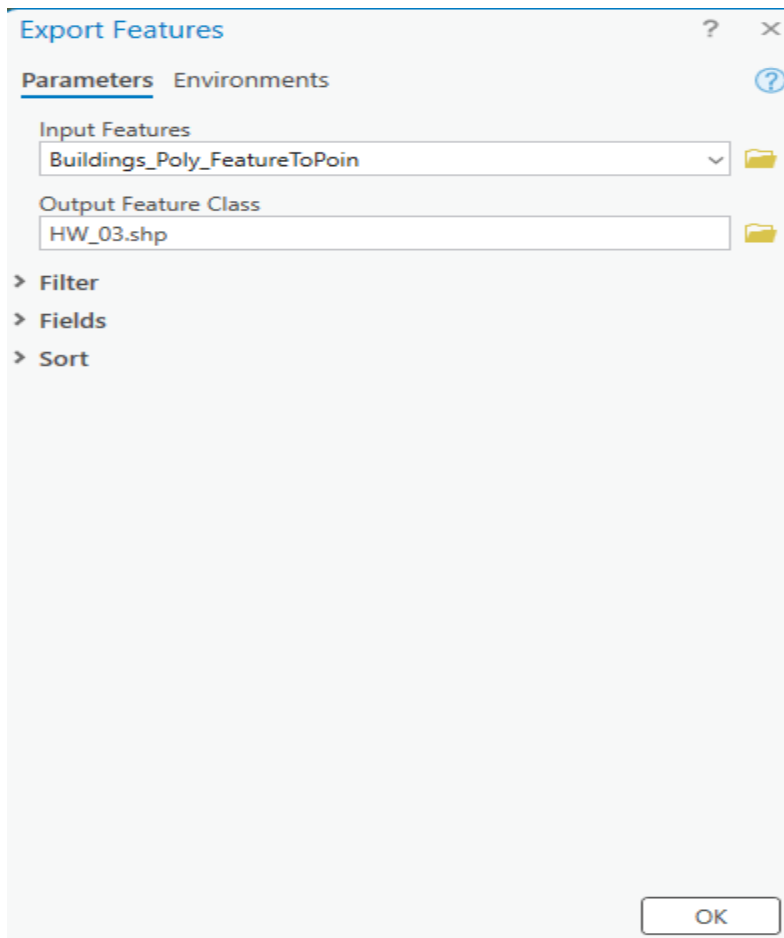
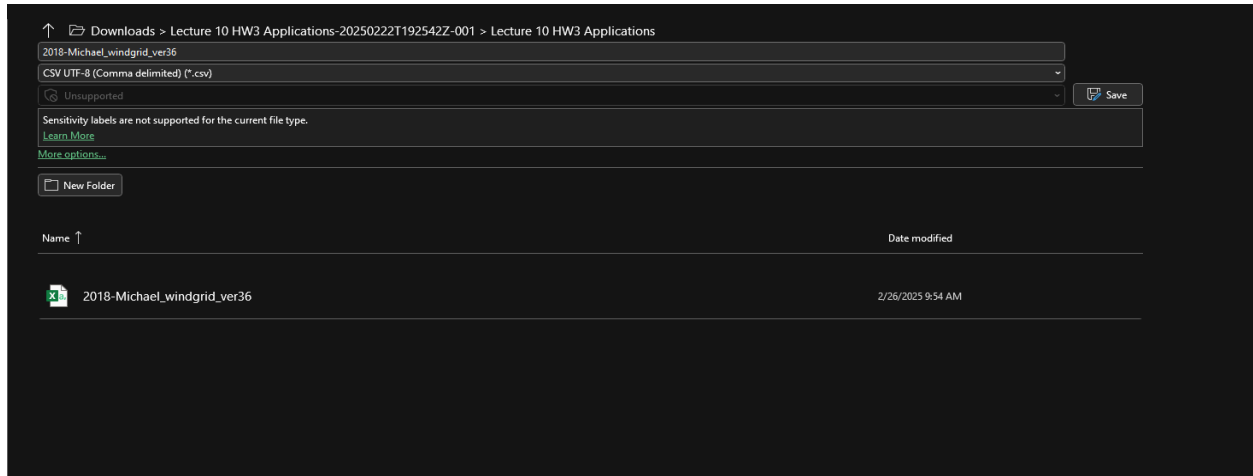
	OBJECTID_1	Shape	OBJECTID	Parcel_ID	Shape_Leng	ORIG_FID	F50yrMaxWSE_WGS84	gulf2016_usace_ncmp_gulf_dem_J1142315_000_000	FD_m	Ele_m
1	1	Point	406279	1375-0000-0351-000	201.701742	4360	2.347943	6.767215	-4.419273	-1.346994
2	2	Point	406280	1375-0000-2003-000	117.379409	4361	2.507159	6.970731	-4.463622	-1.360512
3	3	Point	406281	5914-3000-0012-000	395.980262	4362	2.115684	5.141126	-3.025443	-0.922155
4	4	Point	423180	5914-3000-0011-000	104.651383	10018	2.115684	5.901995	-3.786311	-1.154068
5	5	Point	561427	4130-0000-0000-005	141.422045	26139	2.455186	<Null>	<Null>	<Null>
6	6	Point	561461	4130-0003-0120-000	175.841608	26173	2.481726	2.659901	-0.178176	-0.054308
7	7	Point	561466	4130-0003-0121-000	260.215187	26178	2.46542	6.032941	-3.567521	-1.08738
8	8	Point	561501	4130-0006-0022-000	257.386942	26213	2.496533	4.285805	-1.789272	-0.54537
9	9	Point	561557	4130-0006-0020-000	195.161486	26269	2.514279	5.074435	-2.560175	-0.780341
10	10	Point	561613	4130-0003-0129-000	176.002027	26325	2.514279	4.812633	-2.298373	-0.700544
11	11	Point	561665	4130-0003-0133-000	244.658794	26377	2.549937	4.383416	-1.833479	-0.558844
12	12	Point	561883	4130-0003-0144-000	197.624853	26595	2.618644	4.983234	-2.36461	-0.720733
13	13	Point	561926	4131-0048-0000-000	188.767848	26648	2.774600	2.829530	-0.66409	-0.17116

0 of 29,496 selected Filters: 100%

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Activity 2:

Excel to .csv



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Geoprocessing

XY Table To Point

Parameters

Environments

?

Input Table

2018-Michael_windgrid_ver36.csv

Output Feature Class

HW_03_Raster.shp

X Field

Lon

Y Field

Lat

Z Field

Coordinate System

GCS_WGS_1984

Run

Geoprocessing

←

Point to Raster

+

Parameters

Environments

?

Input Features

HW_03_Raster

Value field

Vg_mph

Output Raster Dataset

Raster_01

Cell assignment type

Most frequent

Priority field

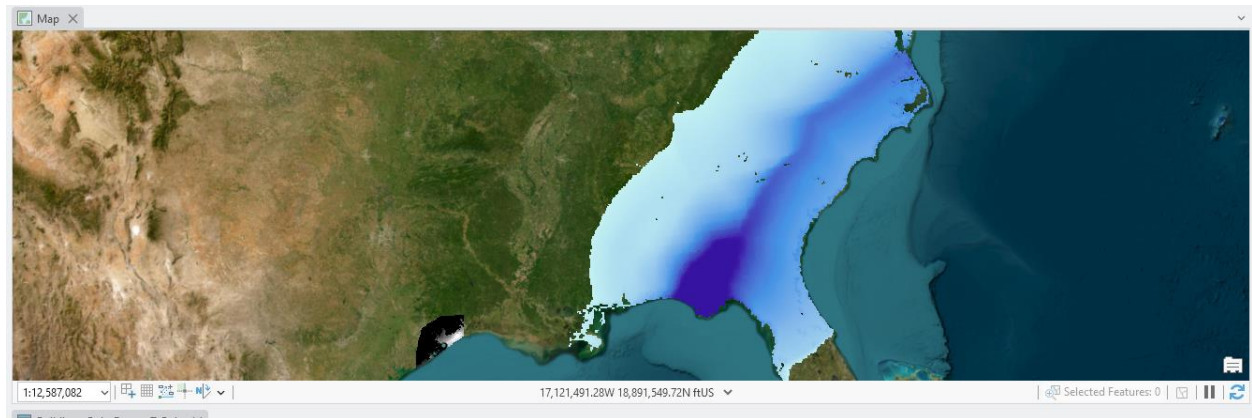
Vg_mph

Cellsize

0.06

☒ Build raster attribute table

▶ Run



MyProject10

Contents

- Map
- HW_03_Raster
- HW_03
- Buildings_Poly_FeatureToPoin
- Buildings_Polygons
- Raster_01
- DEM_1.tif
- DEM.tif
- S01yMaxWSE_WGS84.tif
- World Imagery
- gulf2016_usace_ncomp_gulf_dem_j1142315_00...

Values

157.35

20

52.2853

-31.8545

52.2853

-31.8545

6.05616

0

23.9235

Output Location

Computer > Mostafa Sameer > Downloads > HW_3_6513997

Name	Type	Geometry	Date Modified	Size	Path
info	Folder		2/26/2025 10:35:57 AM		C:\Users\msameer\...
DEM_1.tif	Raster Dataset		2/26/2025 10:20:52 AM	1,364,790 KB	C:\Users\msameer\...
DEM.tif	Raster Dataset		2/26/2025 10:23:40 AM	1,364,790 KB	C:\Users\msameer\...
Raster_01	Raster Dataset		2/26/2025 10:33:57 AM		C:\Users\msameer\...

Name: Raster_6513997

Save Cancel

Symbology - Raster_01

Primary symbology

Stretch

Export Raster

Raster_01

General Settings

Output Raster Dataset

C:\Users\msameer\Documents\ArcGIS\

Output Format

TIFF

Spatial Reference System and Clipping

Coordinate System

GCS_WGS_1984

Geographic Transformations

None

Clipping Geometry

Default

Maintain Clipping Extent

Raster Properties

Cell Size

X: 0.0 Y: 0.0

Raster Size

Columns: 260 Rows: 25

Pixel Type

32 Bit float

NoData value

Renderer Settings

Force RGB

Use Colormap

Use Renderer

Compression Settings

Compression Type

Export

ID	Point	X	Y	Z	Value
3	Point	406281	5914.3000	0012.000	395.900262
4	Point	423180	5914.3000	0011.000	104.651303
5	Point	561427	4130.0000	0000.005	141.422045
6	Point	561461	4130.0003	0120.000	175.841608
7	Point	561466	4130.0003	0121.000	280.215187
8	Point	561501	4130.0006	0022.000	257.386942
9	Point	561557	4130.0006	0020.000	195.161486
10	Point	561613	4130.0003	0129.000	176.002027
11	Point	561665	4130.0003	0133.000	244.658794
12	Point	561883	4130.0003	0144.000	197.624853

0 of 29,496 selected

Export Raster

?

⌵

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✕

Raster_01

General

Settings

Output Raster Dataset

C:\Users\msameer\Downloads\HW_3_65

+

Output Format

TIFF

⌵

▼ Spatial Reference System and Clipping

Coordinate System

GCS_WGS_1984

🌐

Geographic Transformations

None

⌵

Clipping Geometry ⓘ

Default

📁

☐ Maintain Clipping Extent ⓘ

▼ Raster Properties

Cell Size

X

0.0

Y

0.0

Raster Size

Columns

265

Rows

255

Pixel Type

32 Bit float

⌵

NoData value

⌵

▼ Renderer Settings

☐ Force RGB ⓘ

☐ Use Colormap ⓘ

☐ Use Renderer ⓘ

▼ Compression Settings

Compression Type

⌵

Export